

When to Rethink: Learning Multi-Perspective Self-Verification for Vision-Language Models

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Abstract

Vision-language models (VLMs) have achieved strong performance in multimodal reasoning, yet they remain prone to generating plausible but incorrect answers. Self-verification offers a practical way to improve answer reliability without relying on external judges, but existing methods typically depend on a single verification criterion or fixed prompt, resulting in incomplete and unstable reliability estimates. We first systematically analyze how verifier capability and prompt design affect verification performance. Our findings show that stronger verifiers provide more reliable judgments, while verification performance is highly sensitive to prompt choice, with no single prompt consistently dominating across tasks. Guided by these findings, we propose **MOTIVE**, a **Multi-View Self-VerificatiOn wiTh Reliability-Guided SelectiVE** Rethinking framework for reliable multimodal reasoning. **MOTIVE** evaluates each candidate answer from complementary verification perspectives and learns a correctness-aligned reliability score through correctness-grounded multi-view verification learning. During inference, this score governs an accept-or-rethink decision, allowing reliable answers to be returned directly while uncertain ones trigger history-guided rethinking. Extensive experiments across diverse multimodal benchmarks and VLM backbones demonstrate that **MOTIVE** consistently outperforms strong self-verification and self-correction baselines. Further results show that reliable verification improves accept-or-rethink decisions and reduces unnecessary reasoning turns, enabling more reliable and efficient self-verification without an external judge. <https://github.com/misswayguy/Motive>

1 Introduction

Recent advances in vision-language models (VLMs) have demonstrated remarkable capabilities in multimodal understanding, enabling strong performance across diverse visual question answering and reasoning tasks (Dai et al., 2023; Dong and Crisan, 2025; Bordes et al., 2024). However, despite their impressive capabilities, VLMs remain prone to generating plausible but incorrect answers due to visual ambiguity and reasoning errors (Huang et al., 2025; Hou et al., 2024; Chen et al., 2024b). Therefore, enabling VLMs to assess the reliability of their own predictions is crucial for building more trustworthy multimodal systems (Ghosh et al., 2024).

To improve the reliability of VLMs, recent studies have explored self-verification mechanisms that enable models to examine and assess their own generated answers (Zhang et al., 2025b; Chen et al., 2024a; Cheng et al., 2024). Compared with external verification approaches that require additional models or computational

resources, self-verification provides a more efficient way to evaluate predictions within the same multimodal model (Campbell et al., 2024; Anis et al., 2025). However, existing self-verification methods typically rely on a single verification criterion or a fixed checking procedure (Madaan et al., 2023; Shinn et al., 2023), which may fail to capture diverse types of errors in multimodal reasoning. As a result, how to obtain more reliable and comprehensive verification signals remains an open challenge.

To better understand the factors affecting self-verification reliability, we conduct preliminary analyses on verification capability and prompt design in Section 2. Our studies reveal two important observations. ① *Verification effectiveness depends on verifier capability.* Stronger verifiers provide more reliable assessments by better understanding visual evidence, question semantics, and answer consistency, indicating that verification quality is closely related to the reasoning ability of the verifier. ② *Verification performance is sensitive to verification prompt design.* Different verification prompts exhibit complementary strengths, while no single prompt consistently performs best across different multimodal tasks. These findings suggest that reliable self-verification benefits from diverse verification perspectives and the effective aggregation of complementary signals.

Guided by these findings, we introduce MOTIVE (Multi-View Self-VerifiCatiOn wiTh RelIability-Guided SelectiVE Rethinking), a self-verification framework for reliable multimodal reasoning. MOTIVE evaluates each candidate answer through complementary verification views and aggregates their outputs into a unified reliability score. Correctness-grounded verification learning supervises both view-specific judgments and the aggregated score using automatically derived candidate-correctness labels. During inference, the learned score governs an accept-or-rethink decision: reliable answers are returned directly, whereas uncertain ones trigger history-guided rethinking. This design improves verification decisions within the same backbone VLM without requiring an external judge.

Our efforts are unfolded with the following four thrusts:

- ★ **(Preliminary Analysis)** We systematically investigate verification capability and prompt design, showing that stronger verifiers improve answer reliability, while self-verification remains highly prompt-sensitive and no single prompt consistently dominates across tasks.
- ★ **(Algorithm)** Guided by these findings, we propose MOTIVE, a multi-view self-verification framework that uses correctness-grounded verification supervision to support reliability-guided accept-or-rethink decisions without an external judge.
- ★ **(Experiments)** Extensive experiments on five benchmarks and three VLM backbones demonstrate that MOTIVE consistently outperforms strong self-verification and self-correction baselines.
- ★ **(Extra Findings)** We further find that reliable verification enables earlier stopping with fewer unnecessary rethinking turns, while the best individual verification view varies across datasets and backbones, confirming their task-dependent complementarity.

2 Preliminary Analysis

Before presenting our method, we analyze how verification capability and prompt design affect the answer reliability of VLMs in visual question answering.

2.1 Verification Capability

We first examine whether explicit verification improves the answer reliability of VLMs. Given a multimodal query, the baseline directly uses Qwen2.5-VL-3B or Qwen2.5-VL-7B (Bai et al., 2025a) to generate the final answer. In the verification setting, the backbone first produces an initial answer, which is evaluated together with the image and question by an external judge constrained to output YES or NO. A YES verdict

retains the answer, whereas NO triggers rethinking and regeneration. We use InternVL3-2B, 8B, 14B, and 38B (Zhu et al., 2025) as judges and evaluate on VQA (Antol et al., 2015), ScienceQA (Lu et al., 2022), and TableVQA (Kim et al., 2024). The main observations are summarized below.

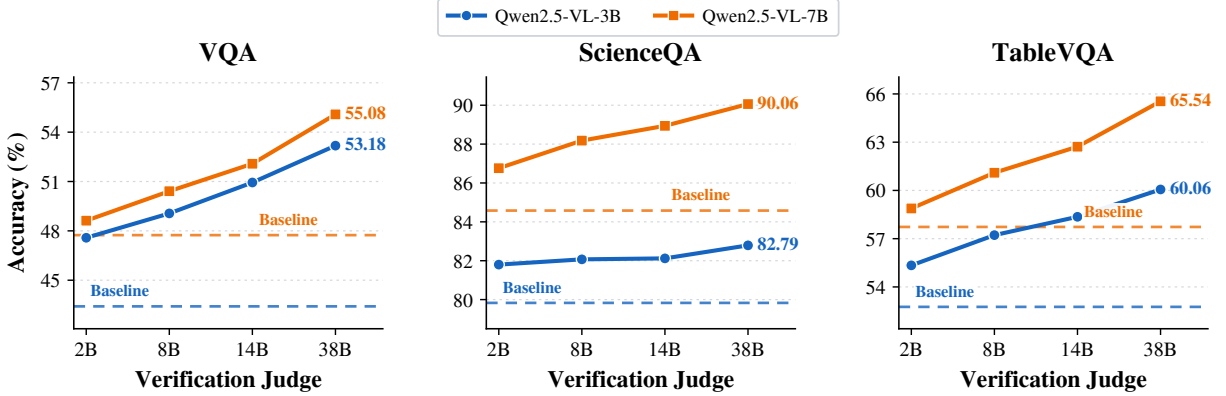


Figure 1: Effect of Verification Judge Capability. Accuracy on three benchmarks using Qwen2.5-VL-3B/7B backbones and InternVL3 judges of different scales. Dashed lines denote performance without verification.

① **Verification Enhances Final Answer Reliability.** As shown in Figure 1, introducing verification consistently improves final-answer performance over direct answering across different backbones and benchmarks. This indicates that the initial responses of VLMs are not always sufficiently reliable. By determining whether an answer should be accepted or regenerated, verification provides an opportunity to correct unreliable responses without modifying or fine-tuning the backbone model.

② **Stronger Judges Improve Verification Effectiveness.** We further observe that larger InternVL3 judges generally achieve better final performance across different Qwen backbones and multimodal benchmarks. This indicates that verification effectiveness depends on the judge’s multimodal reasoning capability, particularly its ability to assess the consistency among visual evidence, question semantics, and the generated answer.

External verification improves VLM reliability but requires an auxiliary judge and introduces additional computational cost, motivating self-verification within the backbone VLM. This provides a more practical way to assess answer reliability without requiring an additional model during use.

2.2 Verification Prompt Design

We next examine how prompt design affects self-verification performance. Since external judges introduce additional computational cost, we use the same backbone VLM for both answer generation and verification. Based on representative self-correction and self-verification protocols (He et al., 2025; Weng et al., 2023; Dhuliawala et al., 2024), we compare two prompt families: five predefined Direct Verification Prompts (DVP-1–DVP-5), provided in Appendix B.1, which assess the initial answer from complementary perspectives, and Generated Verification Questions (GVQ-1–GVQ-3), which generate and answer one, two, or three auxiliary verification questions, respectively, before producing a final YES/NO verdict. A YES verdict retains the initial answer, whereas NO triggers rethinking and regeneration. We conduct the analysis using Qwen2.5-VL-3B on VQA, ScienceQA, and TableVQA, varying only the verification prompt. The main observations are summarized below.

① **Self-Verification Is Highly Prompt-Sensitive.** As shown in Table 1, changing only the verification prompt leads to clear performance variations under the same backbone, datasets, and verification protocol. This indicates that self-verification cannot be reliably elicited by an arbitrary checking instruction. Its effectiveness depends on whether the prompt provides a clear criterion for assessing the initial answer, as different prompts may lead the same VLM to produce inconsistent verification decisions across multimodal tasks.

Table 1: Effect of Verification Prompt Design. Accuracy (%) on VQA, ScienceQA, and TableVQA with Qwen2.5-VL-3B. DVP and GVQ denote direct verification prompts and generated verification questions, respectively. Best results are **bolded**, and second-best results are underlined.

Prompt	VQA	ScienceQA	TableVQA
DVP-1	47.23	81.71	55.00
DVP-2	50.78	<u>81.90</u>	<u>55.67</u>
DVP-3	52.36	82.20	55.33
DVP-4	<u>52.11</u>	81.85	57.00
DVP-5	51.86	81.64	57.00
GVQ-1	51.72	81.36	53.66
GVQ-2	50.64	80.94	52.11
GVQ-3	50.21	80.58	51.44

② **No Single Direct Prompt Consistently Dominates.** Among DVP-1 to DVP-5, the best-performing prompt varies across benchmarks, showing that no single direct prompt provides a universally reliable verification view. Although all DVP prompts follow the same direct YES/NO verification format, their effects are task-dependent, suggesting that each prompt captures only part of the evidence needed for reliable answer checking. Relying on one fixed prompt may therefore overfit the verification behavior to a specific benchmark or question type. This motivates the use of complementary direct prompts to obtain more reliable and informative verification signals across tasks.

③ **Generated Verification Questions Introduce Instability.** GVQ-style prompts are generally less stable than direct verification prompts, and their performance further decreases as more auxiliary questions are generated. Although GVQ makes the verification process more explicit, these intermediate questions are generated by the same VLM that produced the initial answer. As a result, they may inherit the model’s original uncertainty, shift attention away from the original query, or introduce additional reasoning errors before the final verdict. This suggests that increasing the number of self-generated verification steps does not necessarily strengthen verification, but can amplify intermediate uncertainty and error propagation.

These findings indicate that the central challenge of self-verification is not adding more reasoning steps, but establishing a reliable criterion for judging whether the initial answer should be trusted. A single direct prompt provides incomplete and prompt-sensitive signals, while generated verification questions introduce uncertainty. This motivates a multi-view self-verification framework that aggregates complementary signals and learns to distinguish reliable answers from unreliable ones.

3 Method

We propose MOTIVE, a calibrated verification framework for reliable multimodal reasoning, as shown in Figure 2. Given an image–question pair (I, q) , a multimodal policy first generates a candidate answer and then evaluates its reliability through multiple verification views. Instead of treating self-verification as a post-hoc prompting heuristic, MOTIVE learns a correctness-aligned reliability signal that determines whether the model should accept the current answer or allocate additional computation to rethink. The framework jointly optimizes answer generation, correctness-grounded multi-view verification, and reliability-guided selective rethinking.

Problem Setup. Let (I, q, y^*) denote an image, a question, and its ground-truth answer. Given (I, q) , a multimodal policy π_θ generates a candidate answer through an iterative answer–verify–rethink process. At

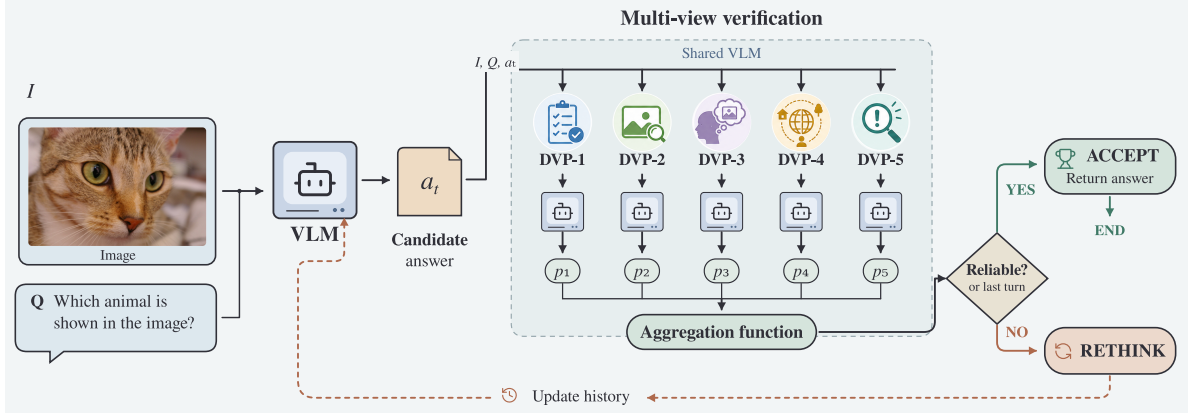


Figure 2: Overview of MOTIVE. The backbone VLM evaluates each candidate answer using five Direct Verification Prompts (DVP-1–DVP-5) and aggregates their scores to guide an accept-or-rethink decision. Accepted answers are returned, while rethinking updates the interaction history and generates a revised candidate.

reasoning turn t , the policy conditions on the interaction history h_t and samples

$$a_t \sim \pi_\theta(\cdot \mid I, q, h_t). \quad (1)$$

The model must then decide whether to return a_t or allocate additional computation to revise it. We formulate this as a binary decision $d_t \in \{\text{accept}, \text{rethink}\}$. If $d_t = \text{accept}$, reasoning terminates and a_t is returned. Otherwise, verification feedback is incorporated into the history and the policy generates a revised candidate at the next turn.

Multi-View Verification. Self-verification can be sensitive to the verification prompt, such that different criteria may yield different judgments for the same candidate answer. Motivated by this observation, we evaluate each candidate from multiple complementary verification views. Let $\mathcal{E}$ denote a set of K verification views. For each $\eta_k \in \mathcal{E}$, we construct a view-specific verification input and estimate the probability that a_t is correct:

$$x_t^{(k)} = \mathcal{V}_{\eta_k}(I, q, h_t, a_t), \quad p_t^{(k)} = \pi_\theta(\text{YES} \mid x_t^{(k)}), \quad k = 1, \dots, K. \quad (2)$$

Here, $\mathcal{V}_{\eta_k}$ specifies the verification instruction associated with the k -th view, and $p_t^{(k)}$ denotes its view-specific reliability estimate. We aggregate these estimates into a unified reliability score

$$s_t = \text{Agg}(p_t^{(1)}, \dots, p_t^{(K)}), \quad (3)$$

which is subsequently used to govern selective rethinking. In our implementation, we use simple averaging for Agg and consider $K = 5$ predefined verification views motivated by the prompt-sensitivity analysis in Section 2.2. The details of multi-view are provided in Appendix B.1.

Proposition 3.1 (Variance reduction from complementary views) *Suppose each verification view provides an unbiased reliability estimate $p_t^{(k)} = \eta_t + \epsilon_t^{(k)}$, with $\text{Var}(\epsilon_t^{(k)}) \leq \sigma^2$ and pairwise $\text{Cov}(\epsilon_t^{(j)}, \epsilon_t^{(k)}) \leq \rho\sigma^2$ for $j \neq k$. For mean aggregation, $s_t = \frac{1}{K} \sum_{k=1}^K p_t^{(k)}$,*

$$\mathbb{E}[(s_t - \eta_t)^2] \leq \sigma^2 \left(\rho + \frac{1 - \rho}{K} \right).$$

Takeaway. The bound decreases with K whenever $\rho < 1$. Thus, averaging provides greater variance reduction when verification errors are less correlated, formally motivating the use of complementary rather than redundant views.

Correctness-Grounded Multi-View Verification Learning. Outcome-level rewards supervise the final answer but do not explicitly teach the model whether an intermediate candidate should be trusted. We therefore introduce an auxiliary verification objective that aligns both view-specific and aggregated reliability estimates with candidate correctness. For candidate $a_{i,t}$ at turn t of rollout o_i , we define its correctness label as $z_{i,t} = M(a_{i,t}, y^*) \in \{0, 1\}$, where M is the task-specific correctness function. Using binary cross-entropy $\ell_{\text{BCE}}(z, p) = -z \log p - (1 - z) \log(1 - p)$, we first supervise each verification view independently:

$$\mathcal{L}_{\text{view}} = \frac{1}{G} \sum_{i=1}^G \frac{1}{T_i} \sum_{t=1}^{T_i} \frac{1}{K} \sum_{k=1}^K \ell_{\text{BCE}}(z_{i,t}, p_{i,t}^{(k)}). \quad (4)$$

Here, G denotes the number of rollouts and T_i the number of candidate-answer turns in rollout o_i . This view-level supervision discourages individual views from becoming uninformative and encourages each to provide a correctness-sensitive verification signal. We further align the aggregated reliability score with candidate correctness:

$$\mathcal{L}_{\text{agg}} = \frac{1}{G} \sum_{i=1}^G \frac{1}{T_i} \sum_{t=1}^{T_i} \ell_{\text{BCE}}(z_{i,t}, s_{i,t}). \quad (5)$$

Because $s_{i,t}$ is subsequently used as a continuous decision signal, its probability magnitude is also important: two predictions with the same binary decision can imply substantially different accept-or-rethink behavior. We therefore complement the log-loss objective with a Brier-style squared-error regularizer,

$$\mathcal{L}_{\text{cal}} = \frac{1}{G} \sum_{i=1}^G \frac{1}{T_i} \sum_{t=1}^{T_i} (s_{i,t} - z_{i,t})^2, \quad (6)$$

which directly penalizes deviations between the reliability score and the observed correctness outcome. Rather than replacing the probabilistic supervision provided by binary cross-entropy, this term provides a complementary constraint on the numerical scale of $s_{i,t}$, making it better suited for threshold-based selective inference. The resulting verification objective is

$$\mathcal{L}_{\text{ver}} = \mathcal{L}_{\text{view}} + \beta_{\text{agg}} \mathcal{L}_{\text{agg}} + \beta_{\text{cal}} \mathcal{L}_{\text{cal}}, \quad (7)$$

where β_{agg} and β_{cal} control the contributions of aggregate-level correctness alignment and reliability-score regularization, respectively.

Reliability-Guided Selective Rethinking. The calibrated reliability score provides a stopping criterion for inference-time reasoning. The initial prompt is shown in Appendix B.2. At turn t , after generating a candidate answer a_t , MOTIVE evaluates it under K verification views and computes the aggregated reliability score s_t . The model accepts the current answer when its reliability score reaches the threshold τ ; otherwise, another rethinking step is performed unless the maximum reasoning budget has been reached:

$$d_t = \begin{cases} \text{accept}, & s_t \geq \tau \text{ or } t = T, \\ \text{rethink}, & s_t < \tau \text{ and } t < T. \end{cases} \quad (8)$$

If no candidate reaches the acceptance threshold before the maximum reasoning budget is exhausted, the final candidate answer a_T is returned.

When rethinking is triggered, the candidate answer, the verification outputs, and a rethinking instruction are appended to the history:

$$h_{t+1} = \text{Update}(h_t, a_t, v_t^{(1)}, \dots, v_t^{(K)}, u_{\text{rethink}}), \quad (9)$$

where $v_t^{(k)}$ denotes the binary verification output from the k -th view, and u_{rethink} instructs the model to reconsider the image, question, and previous answer. The updated history h_{t+1} is then used to generate a revised answer a_{t+1} .

To discourage unnecessary rethinking, we include a computation-aware reward:

$$\tilde{r}_i = M\left(a_{\text{sel}}^{(i)}, y^*\right) - c \sum_{t=1}^{T_i} \mathbb{I}[d_{i,t} = \text{rethink}], \quad (10)$$

where c is the computational penalty associated with an additional rethinking turn. This encourages the model to improve final-answer correctness while avoiding excessive computation.

Overall Objective. The final training objective combines outcome-level answer optimization with correctness-grounded multi-view verification:

$$\mathcal{L}_{\text{MOTIVE}} = \mathcal{L}_{\text{GRPO}} + \lambda_{\text{ver}} \mathcal{L}_{\text{ver}}, \quad (11)$$

where λ_{ver} balances answer-generation learning and verifier supervision. In practice, $\mathcal{L}_{\text{GRPO}}$ is applied only to answer-generation tokens, while $\mathcal{L}_{\text{ver}}$ is applied to verification outputs. This separates final-answer optimization from reliability estimation, while still allowing the verifier to guide rethinking behavior during inference.

4 Experiments

Datasets. We conduct experiments on five visual question answering benchmarks: ReFocus (Fu et al., 2025), VQA (Antol et al., 2015), SLaKE (Liu et al., 2021), ScienceQA (Lu et al., 2022), and TableVQA (Kim et al., 2024). The selected benchmarks cover general image understanding, structured-image reasoning, medical VQA, scientific multimodal reasoning, and table-based visual question answering. Specifically, VQA evaluates general visual question answering over natural images; ReFocus focuses on structured visual inputs such as charts and tables; SLaKE targets medical VQA with radiology images; ScienceQA requires image-grounded scientific reasoning; and TableVQA tests fine-grained understanding of table images. We follow the official splits and evaluation protocols for all datasets. More dataset details are provided in Appendix A.1.

Models and Baselines. We evaluate MOTIVE on three representative publicly available VLM backbones: Qwen2.5-VL-3B, Qwen2.5-VL-7B (Bai et al., 2025a), and InternVL3-2B (Zhu et al., 2025). These models cover different parameter scales and model families, allowing us to evaluate whether MOTIVE consistently improves verification reliability across diverse vision-language backbones rather than being specific to a particular model architecture. We compare MOTIVE with several groups of baselines. The first baseline is the vanilla VLM, which directly generates the final answer without self-verification. The second group includes prompt-level self-reasoning and self-verification methods, including Self-Consistency (Wang et al., 2023), Self-Verification (Weng et al., 2023), and Chain-of-Verification (Dhuliawala et al., 2024). The third group includes recent multimodal self-improvement and self-correction methods, including ADPO (Qiu et al., 2026), R³V (Cheng et al., 2025), and LLaVA-SCo (Liu et al., 2025). For methods originally developed for language-only reasoning, we adapt their verification or correction prompts to the vision-language setting by conditioning on the same image-question input. All baselines are evaluated under the same datasets, answer-matching protocol, and decoding budget unless otherwise specified.

Implementation Details. For each backbone, all methods are evaluated using the same dataset splits and answer-matching protocol. We train MOTIVE with the AdamW optimizer using an initial learning rate of 1×10^{-6} and BF16 precision. During GRPO training, we use a rollout group size of $G = 8$ and a sampling temperature of 1.0. We employ $K = 5$ verification views and aggregate their reliability scores by arithmetic averaging. The acceptance threshold is set to $\tau = 0.5$, the maximum number of candidate-answer turns is $T = 3$, and the verification objective is weighted by $\lambda_{\text{ver}} = 0.1$. We use official evaluation scripts when available; otherwise, multiple-choice and open-form answers are evaluated by exact option matching and normalized answer matching, respectively. Additional hyperparameters are provided in Appendix A.2.

4.1 Main Results

Table 2 summarizes the main comparison across five VQA benchmarks on three VLM backbones. We observe that prompt-level methods such as Self-Consistency, Self-Verification, and CoVe improve over the vanilla VLM, but their gains remain limited, suggesting that prompt-only self-checking is not sufficient for reliable visual answer verification. Recent multimodal self-improvement methods, including ADPO, R³V, and LLaVA-SCo, provide stronger baselines, yet MOTIVE consistently achieves the best performance across different model scales and task domains. This consistent pattern indicates that the advantage of MOTIVE comes from modeling and aggregating complementary verification signals, rather than relying solely on sampling, reflection, or post-hoc correction.

Table 2: Main results across five VQA benchmarks. Top-1 accuracy (%) is reported using three VLM backbones on five VQA benchmarks and the average score. Best results are **bolded**, and second-best results are underlined.

Backbone	Method	ReFocus	VQA	SLaKE	ScienceQA	TableVQA	Average
Qwen2.5-VL-3B	Baseline	51.63	43.41	47.91	79.83	52.76	55.11
	Self-Cons. (Wang et al., 2023)	52.31	44.16	48.62	81.09	53.41	55.92
	Self-Verif. (Weng et al., 2023)	53.62	47.68	49.51	81.76	55.12	57.54
	CoVe (Dhuliawala et al., 2024)	53.87	47.41	49.88	81.79	54.36	57.46
	ADPO (Qiu et al., 2026)	<u>56.08</u>	51.48	<u>54.21</u>	<u>82.35</u>	<u>57.91</u>	<u>60.41</u>
	R ³ V (Cheng et al., 2025)	55.74	<u>51.62</u>	53.46	82.29	57.48	60.12
	LLaVA-SCo (Liu et al., 2025)	55.21	50.91	52.73	82.12	56.92	59.58
	MOTIVE (Ours)	58.65	54.22	58.75	83.18	60.85	63.13
Qwen2.5-VL-7B	Baseline	57.62	47.74	54.77	84.58	57.73	60.49
	Self-Cons. (Wang et al., 2023)	58.39	48.58	55.48	85.07	58.46	61.20
	Self-Verif. (Weng et al., 2023)	61.84	49.48	59.16	87.72	59.91	63.62
	CoVe (Dhuliawala et al., 2024)	60.73	51.74	58.72	87.36	58.64	63.44
	ADPO (Qiu et al., 2026)	63.32	53.06	<u>61.96</u>	88.47	<u>63.24</u>	<u>66.01</u>
	R ³ V (Cheng et al., 2025)	63.84	53.28	61.73	<u>88.91</u>	61.74	65.90
	LLaVA-SCo (Liu et al., 2025)	63.21	52.64	61.26	88.62	62.57	65.66
	MOTIVE (Ours)	66.45	56.10	63.72	90.68	66.20	68.63
InternVL3-2B	Baseline	53.42	44.86	49.68	81.36	53.18	56.50
	Self-Cons. (Wang et al., 2023)	54.11	45.63	50.26	81.91	53.92	57.17
	Self-Verif. (Weng et al., 2023)	56.21	48.36	52.41	83.24	56.74	59.39
	CoVe (Dhuliawala et al., 2024)	55.86	48.12	52.03	83.06	55.91	58.99
	ADPO (Qiu et al., 2026)	<u>58.42</u>	52.04	<u>56.17</u>	84.39	<u>59.82</u>	<u>62.17</u>
	R ³ V (Cheng et al., 2025)	58.13	<u>52.28</u>	55.84	<u>84.51</u>	59.36	62.02
	LLaVA-SCo (Liu et al., 2025)	57.64	51.58	55.12	84.16	58.74	61.45
	MOTIVE (Ours)	61.75	55.08	60.05	86.22	62.80	65.18

4.2 Reliability of Accept-or-Rethink Decisions

To examine whether the learned reliability score can correctly determine when a candidate answer should be accepted or reconsidered, we evaluate the accept-or-rethink decisions using Self-Verification Accuracy (SVA). A decision is regarded as correct when the model accepts a correct candidate answer or sends an incorrect candidate answer for rethinking. We compare conventional Self-Verification with MOTIVE across five benchmarks using three VLM backbones. As shown in Table 3, MOTIVE consistently achieves higher SVA across all evaluated backbones and datasets, with particularly clear improvements on tasks requiring fine-grained visual grounding and structured evidence checking. The gains are also maintained when the conventional self-verifier already performs strongly, demonstrating that the aggregated reliability score provides a more dependable signal for deciding whether the current answer should be retained or further revised.

Table 3: Self-Verification Accuracy (SVA %) for accept-or-rethink decisions. Results are based on five benchmarks using three VLM backbones. Higher values indicate more accurate verification decisions, and the best results are shown in **bold**.

Backbone	Method	ReFocus	VQA	SLaKE	ScienceQA	TableVQA	Average
Qwen2.5-VL-3B	Self-Verification	66.85	64.20	63.90	82.85	66.75	68.91
	MOTIVE	75.10	74.00	76.80	86.05	75.45	77.48
Qwen2.5-VL-7B	Self-Verification	71.20	65.60	69.40	86.10	69.30	72.32
	MOTIVE	78.65	75.10	76.40	90.10	78.20	79.69
InternVL3-2B	Self-Verification	67.80	64.75	66.10	84.20	67.50	70.07
	MOTIVE	76.25	74.55	76.60	88.45	76.65	78.50

4.3 Effectiveness of Multi-View Verification

To verify that the gains of MOTIVE arise from combining complementary verification views rather than from a single strong prompt, we evaluate five single-view variants and the full model on five benchmarks with three VLM backbones. Each variant retains one verification view while all other settings remain unchanged. As shown in Table 4, the best single view changes across datasets and backbones, and no individual view is consistently superior. In contrast, MOTIVE achieves the highest and most stable overall accuracy, validating the necessity of multi-view verification.

4.4 Efficient Reasoning via Reliable Early Stopping

To examine whether MOTIVE can avoid unnecessary rethinking, we compare its average reasoning turns with conventional Self-Verification under the same maximum budget of $T = 3$. A lower value indicates that the model can accept reliable answers earlier and requires fewer repeated revisions. As shown in Table 5, MOTIVE consistently reduces the average number of reasoning turns across all backbones and benchmarks. The reduction is more pronounced on challenging tasks such as VQA and SLaKE, where conventional Self-Verification is more likely to repeatedly reconsider uncertain answers, while smaller reductions are observed on ScienceQA, where reliable answers can already be identified relatively early. These results demonstrate that MOTIVE not only improves answer reliability but also enables more efficient reasoning through better early-stopping decisions.

4.5 Effect of Prompt Redundancy and Diversity

We investigate how the overlap and diversity among verification prompts affect multi-view verification using Qwen2.5-VL-3B across five benchmarks. Setting 1 repeats DVP-1 in all slots, representing maximally

Table 4: Complementarity of individual verification views. Accuracy (%) of five single-view variants and the complete MOTIVE aggregation on five benchmarks using three VLM backbones. The best overall result is shown in **bold**, and the best single-view result is underlined.

Backbone	Setting	ReFocus	VQA	SLaKE	ScienceQA	TableVQA	Average
Qwen2.5-VL-3B	DVP-1	54.20	47.23	52.80	81.71	55.00	58.19
	DVP-2	<u>57.10</u>	50.78	55.60	81.90	55.67	60.21
	DVP-3	55.80	<u>52.36</u>	56.40	<u>82.20</u>	55.33	60.42
	DVP-4	56.30	52.11	55.90	81.85	<u>57.00</u>	60.63
	DVP-5	56.70	51.86	<u>57.20</u>	81.64	<u>57.00</u>	60.88
	MOTIVE	58.65	54.22	58.75	83.18	60.85	63.13
Qwen2.5-VL-7B	DVP-1	61.90	50.20	59.10	89.30	61.60	64.42
	DVP-2	63.80	53.40	61.10	<u>89.95</u>	62.70	66.19
	DVP-3	62.90	<u>54.80</u>	62.20	89.60	62.10	66.32
	DVP-4	<u>65.20</u>	54.20	61.60	89.10	<u>64.30</u>	<u>66.88</u>
	DVP-5	64.10	53.90	<u>62.90</u>	89.35	63.90	66.83
	MOTIVE	66.45	56.10	63.72	90.68	66.20	68.63
InternVL3-2B	DVP-1	57.30	49.60	55.80	84.90	58.60	61.24
	DVP-2	59.20	52.10	57.40	<u>85.55</u>	59.80	62.81
	DVP-3	58.10	<u>53.85</u>	58.20	85.20	59.60	62.99
	DVP-4	59.70	53.10	<u>59.00</u>	84.95	<u>60.95</u>	<u>63.54</u>
	DVP-5	<u>60.40</u>	52.70	58.70	85.10	60.60	63.50
	MOTIVE	61.75	55.08	60.05	86.22	62.80	65.18

Table 5: Average reasoning turns under a maximum budget of $T = 3$. Results compare conventional Self-Verification and MOTIVE on five benchmarks using three VLM backbones. Lower values are better, and the best results are shown in **bold**.

Backbone	Method	ReFocus ↓	VQA ↓	SLaKE ↓	ScienceQA ↓	TableVQA ↓	Average ↓
Qwen2.5-VL-3B	Self-Verification	2.41	2.88	2.74	2.32	2.57	2.58
	MOTIVE	2.13	2.49	2.40	2.14	2.30	2.29
Qwen2.5-VL-7B	Self-Verification	2.34	2.72	2.59	2.21	2.45	2.46
	MOTIVE	2.10	2.41	2.31	2.09	2.23	2.23
InternVL3-2B	Self-Verification	2.39	2.80	2.67	2.26	2.51	2.53
	MOTIVE	2.12	2.44	2.35	2.11	2.26	2.26

redundant verification signals; Settings 2–4 successively replace repeated copies of DVP-1 with DVP-2, DVP-3, and DVP-4; and Setting 5 uses the complete set of DVP-1–DVP-5. As shown in Figure 3, performance generally improves as repeated prompts are replaced by distinct verification views, with the complete configuration achieving the best overall result despite minor task-specific fluctuations. This pattern indicates that repeated prompts provide largely overlapping information, whereas prompts based on different criteria contribute complementary evidence; the gradually saturating improvement also suggests that the newly introduced views retain some shared verification information.

4.6 Ablation Study

Verification Loss Components. We progressively add $\mathcal{L}_{\text{view}}$, $\mathcal{L}_{\text{agg}}$, and $\mathcal{L}_{\text{cal}}$ to the GRPO objective while keeping all other settings unchanged. As shown in Table 6, $\mathcal{L}_{\text{view}}$ provides the largest improvement by directly supervising individual verification views, while $\mathcal{L}_{\text{agg}}$ further improves the aggregated reliability score

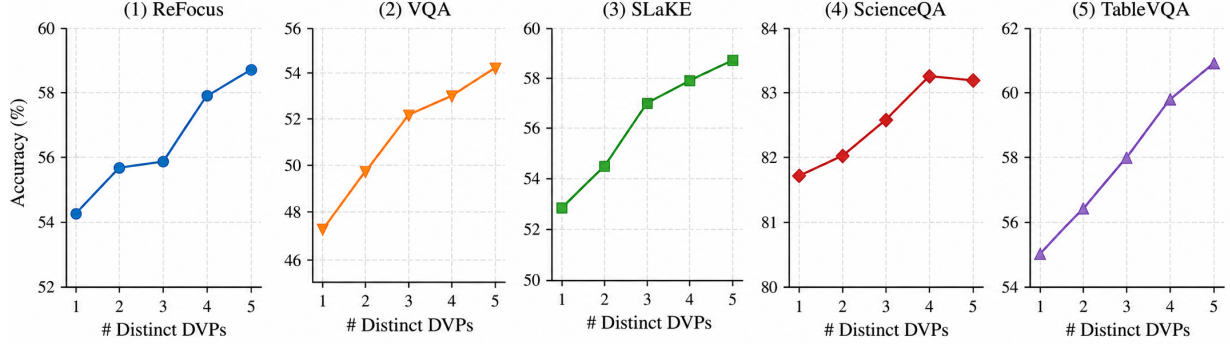


Figure 3: Effect of verification-view diversity. Accuracy (%) on five benchmarks as the number of distinct verification prompts increases from one to five, while keeping five verification slots.

used for accept-or-rethink decisions. $\mathcal{L}_{\text{cal}}$ provides an additional improvement by regularizing the numerical scale of the aggregated score. Combining all three components achieves the best overall performance.

Table 6: Ablation of verification loss components. Final answer accuracy (%) is reported on five benchmarks using three VLM backbones. The best results are shown in **bold**.

Backbone	Training Objective	ReFocus	VQA	SLaKE	ScienceQA	TableVQA	Average
Qwen2.5-VL-3B	$\mathcal{L}_{\text{GRPO}}$	51.63	43.41	47.91	79.83	52.76	55.11
	$+\mathcal{L}_{\text{view}}$	56.61	50.54	55.28	82.24	58.34	60.60
	$+\mathcal{L}_{\text{view}} + \mathcal{L}_{\text{agg}}$	58.16	53.14	57.77	82.91	60.12	62.42
	$+\mathcal{L}_{\text{view}} + \mathcal{L}_{\text{agg}} + \mathcal{L}_{\text{cal}}$	58.65	54.22	58.75	83.18	60.85	63.13
Qwen2.5-VL-7B	$\mathcal{L}_{\text{GRPO}}$	57.62	47.74	54.77	84.58	57.73	60.49
	$+\mathcal{L}_{\text{view}}$	63.71	53.42	61.03	88.97	63.57	66.14
	$+\mathcal{L}_{\text{view}} + \mathcal{L}_{\text{agg}}$	65.74	55.35	63.00	90.25	65.44	67.96
	$+\mathcal{L}_{\text{view}} + \mathcal{L}_{\text{agg}} + \mathcal{L}_{\text{cal}}$	66.45	56.10	63.72	90.68	66.20	68.63
InternVL3-2B	$\mathcal{L}_{\text{GRPO}}$	53.42	44.86	49.68	81.36	53.18	56.50
	$+\mathcal{L}_{\text{view}}$	59.17	51.71	56.73	84.86	59.82	62.46
	$+\mathcal{L}_{\text{view}} + \mathcal{L}_{\text{agg}}$	61.08	54.16	59.22	85.88	61.93	64.45
	$+\mathcal{L}_{\text{view}} + \mathcal{L}_{\text{agg}} + \mathcal{L}_{\text{cal}}$	61.75	55.08	60.05	86.22	62.80	65.18

Reliability Score Aggregation. We investigate the effect of different aggregation strategies for combining multiple verification views while keeping all other training and inference settings unchanged. Specifically, we compare Majority Vote, which first converts each verification view into a binary decision and then selects the majority prediction, with Averaging, which directly aggregates the continuous YES probabilities produced by all verification views. As shown in Table 7, Averaging consistently achieves better performance than Majority Vote across different backbones and benchmarks. This indicates that preserving the confidence information contained in each verification view provides a more reliable estimation of answer correctness than using only discrete verification decisions.

Verification Loss Weight. We investigate the effect of the verification loss weight λ_{ver} by varying its value while keeping other settings unchanged. As shown in Figure 4, removing the verification objective ($\lambda_{\text{ver}} = 0$) leads to a clear performance drop, confirming the importance of verification supervision. The best performance is achieved at $\lambda_{\text{ver}} = 0.1$, while larger weights slightly degrade the results, suggesting that excessive verification supervision may affect answer generation.

Aggregate-Level Verification Loss Weight. We further study the effect of the aggregate-level verification loss weight β_{agg} while keeping all other settings unchanged. As shown in Figure 5, removing aggregate-level supervision decreases performance, indicating that explicitly aligning the aggregated reliability score with

Table 7: Ablation of reliability score aggregation strategies. Final answer accuracy (%) is reported on five benchmarks using three VLM backbones. The best results are shown in **bold**.

Backbone	Aggregation	ReFocus	VQA	SLaKE	ScienceQA	TableVQA	Average
Qwen2.5-VL-3B	Majority Vote	58.21	53.71	58.31	82.87	60.29	62.68
	Averaging	58.65	54.22	58.75	83.18	60.85	63.13
Qwen2.5-VL-7B	Majority Vote	65.88	55.74	63.35	90.41	65.71	68.22
	Averaging	66.45	56.10	63.72	90.68	66.20	68.63
InternVL3-2B	Majority Vote	61.36	54.63	59.68	85.86	62.39	64.78
	Averaging	61.75	55.08	60.05	86.22	62.80	65.18

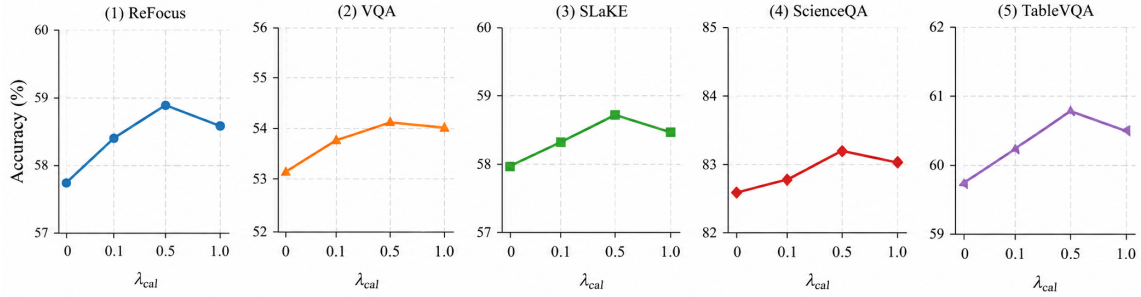


Figure 4: Effect of the verification loss weight λ_{ver} . Accuracy (%) on five benchmarks under different values of λ_{ver} using Qwen2.5-VL-3B.

candidate correctness improves accept-or-rethink decisions. The best results are obtained with $\beta_{agg} = 1.0$, whereas larger values slightly reduce final-answer accuracy.

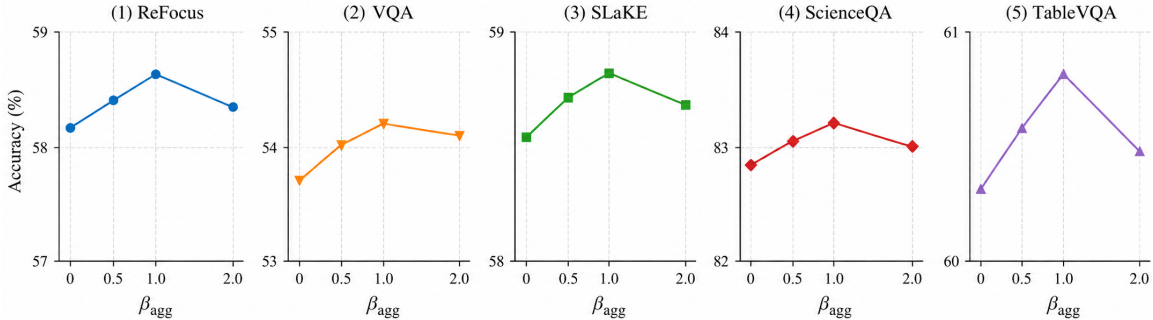


Figure 5: Effect of the aggregate-level verification loss weight β_{agg} . Accuracy (%) on five benchmarks under different values of β_{agg} using Qwen2.5-VL-3B.

Calibration Loss Weight. We further analyze the effect of the calibration loss weight β_{cal} while fixing the other loss weights. As shown in Figure 6, removing $\mathcal{L}_{cal}$ slightly decreases final-answer accuracy, suggesting that regularizing the aggregated reliability score benefits selective verification. The best results are obtained with $\beta_{cal} = 0.1$, while larger weights provide limited benefits and slightly affect the final accuracy.

5 Related Work

Self-Verification in Multimodal Reasoning. Recent advances in multimodal reasoning have increasingly explored self-verification and self-correction mechanisms to improve the reliability of vision-language models (Ding et al., 2026). Rather than relying solely on a single-pass prediction, these methods encourage

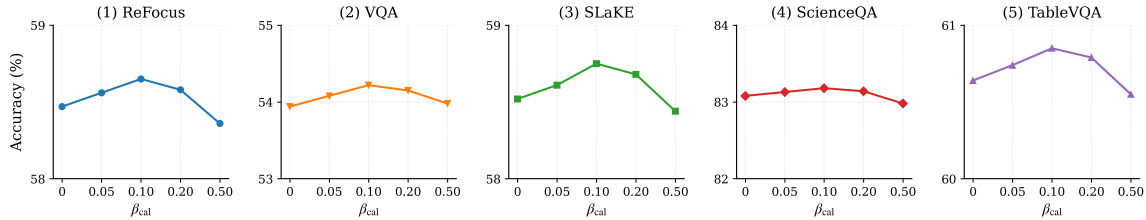


Figure 6: Effect of the calibration loss weight β_{cal} . Accuracy (%) on five benchmarks under different values of β_{cal} using Qwen2.5-VL-3B.

models to re-examine their intermediate reasoning or final answers through reflection (Yan et al., 2024; Cao et al., 2025; Zhang et al., 2026; Sun et al., 2025; Wang et al., 2025a), critique (Madaan et al., 2023; Shinn et al., 2023; Wu et al., 2025b), verifier-guided assessment (Weng et al., 2023; Ding and Zhang, 2025; Wu et al., 2025a), or iterative correct (Gou et al., 2024; Manakul et al., 2023; Cheng et al., 2025; He et al., 2025; Liu et al., 2025). Such designs have shown promise in reducing inconsistent predictions and improving performance on complex visual reasoning tasks (Qian et al., 2026; Lee et al., 2024), where errors often arise from weak visual grounding or unstable multi-step inference (Yin et al., 2024; Si et al., 2021). However, most existing approaches treat verification primarily as an inference-time strategy for refining final outputs (Luo et al., 2025; Li et al., 2026a), instead of explicitly modeling the supervision carried by the verification process itself. As a result, the fine-grained signals embedded in multi-turn self-verification, especially those associated with intermediate candidate answers, remain underexploited during training.

Verifiable Supervision and Training Signals for VLMs. Beyond inference-time correction, another line of research seeks to improve reasoning models by turning verifiable outcomes into explicit training signals (Cobbe et al., 2021; Lightman et al., 2023; Shao et al., 2024; Wu et al., 2026b,a). In tasks where answer correctness can be automatically checked, prior work has explored reward-based learning (Bai et al., 2025b; Meng et al., 2025), process supervision (Zhang et al., 2025a; Li et al., 2026b; Han et al., 2026), and intermediate guidance to enhance model reliability without relying entirely on manual annotations (Guo et al., 2025; Zhang et al., 2025b; Zhan et al., 2025; Shen et al., 2025). These methods demonstrate that verifiable feedback can serve as an effective and scalable supervision source for vision-language models, especially in settings that require structured reasoning and precise decision making (Chen and Song, 2025; Marsili and Gkioxari, 2025; Zhou et al., 2025). Nevertheless, existing approaches typically emphasize final-answer correctness (Wang et al., 2025b; Ong et al., 2025; Zhou et al., 2024), while paying less attention to the supervision value of intermediate verification trajectories. In particular, they seldom model the consistency between round-wise candidate answers and corresponding verification judgments, leaving multi-turn self-verification signals insufficiently exploited during training.

6 Conclusion

In this work, we investigated self-verification in vision-language models and showed that its effectiveness depends on verifier capability and prompt design, with no single verification prompt consistently dominating across tasks. Guided by these findings, we proposed MOTIVE, a multi-view self-verification framework that aggregates complementary verification signals and learns correctness-aligned reliability through view-level and aggregate-level verification supervision. The resulting reliability score supports accept-or-rethink decisions, enabling uncertain answers to be selectively revised within the same backbone VLM without requiring an external judge. Experiments across five benchmarks and three VLM backbones demonstrate that MOTIVE improves final-answer performance, verification decision quality, and early stopping.

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A Experiment Settings

A.1 Additional Dataset Details

ReFocus. ReFocus (Fu et al., 2025) evaluates structured-image understanding over visually complex tables and charts. The benchmark contains 2,219 question–answer pairs collected from six subsets covering Wikipedia tables, synthetically rendered tables, table verification, scientific charts, and horizontal and vertical bar charts. Its questions require fine-grained information extraction, comparison, counting, aggregation, and multi-step reasoning over structured visual content.

VQA. VQA (Antol et al., 2015) is a large-scale open-ended visual question answering benchmark based primarily on natural images from Microsoft COCO. Its real-image portion contains approximately 204,000 images and 614,000 human-written questions covering object recognition, attributes, counting, spatial relationships, activities, and commonsense reasoning. Multiple human answers are collected for each question to account for linguistic variation and answer ambiguity.

SLaKE. SLaKE (Liu et al., 2021) is a bilingual medical VQA dataset containing 642 radiological images and 14,028 English–Chinese question–answer pairs. It includes CT, MRI, and X-ray images covering the head, neck, chest, abdomen, and pelvic cavity. The questions examine imaging modality, anatomical structures, organ positions, abnormalities, and medical knowledge. We use the English-language questions in our experiments.

ScienceQA. ScienceQA (Lu et al., 2022) contains 21,208 multiple-choice questions collected from elementary- and high-school science curricula, of which 10,332 are associated with image contexts. The questions span natural science, social science, and language science and may include scientific diagrams, natural images, textual contexts, candidate answers, lectures, and explanations. We use the image-based subset to evaluate multimodal scientific reasoning.

TableVQA. TableVQA-Bench (Kim et al., 2024) contains 1,500 question–answer pairs constructed from table images across multiple domains, including Wikipedia and financial documents. The tables exhibit diverse layouts, styles, fonts, borders, and cell structures. The benchmark evaluates visual cell recognition, row–column alignment, comparison, aggregation, numerical calculation, and reasoning over structured table content.

A.2 Additional Implementation Details

All experiments are conducted on NVIDIA B200 GPUs.¹ The internal weights of the verification objective are set to $\beta_{\text{agg}} = 1.0$ and $\beta_{\text{cal}} = 0.1$, while the computation-aware reward coefficient is set to $c = 0.05$. The verification loss weight is set to $\lambda_{\text{ver}} = 0.1$. During inference, MOTIVE uses $K = 5$ verification views, an acceptance threshold of $\tau = 0.5$, and a maximum of $T = 3$ reasoning turns. For each verification view, the view-specific reliability estimate is given by the probability assigned to the YES output, and the final reliability score is obtained by averaging the estimates across all verification views. The computation-aware rollout reward is used to compute group-relative advantages in GRPO, while the auxiliary verification objective jointly supervises view-specific judgments and aggregated reliability scores.

¹All experiments, data and model access, and processing activities were conducted by the University of Leicester.

B Prompt Details

B.1 Direct Verification Prompt Templates

In Section 2.2, we analyze different verification prompt designs and observe that no single direct verification prompt consistently dominates across all benchmarks. Based on this analysis, we adopt the five predefined Direct Verification Prompts (DVPs) as complementary verification views in MOTIVE, as described in Section 3. These prompts evaluate a candidate answer from different perspectives, covering overall correctness, visual evidence support, backward evidence checking, global contextual consistency, and error discovery. Specifically, DVP-1 performs general correctness verification without additional constraints; DVP-2 focuses on whether the answer is directly grounded in observable image evidence; DVP-3 verifies whether the evidence required by the answer can be identified by reasoning back from the predicted conclusion; DVP-4 examines the consistency between the answer, the question context, and the overall visual scene; and DVP-5 adopts an error-seeking perspective to detect potential contradictions or mistakes in the candidate answer. Together, these complementary views provide diverse criteria for evaluating answer reliability. The complete DVP templates are summarized in Figure 7.

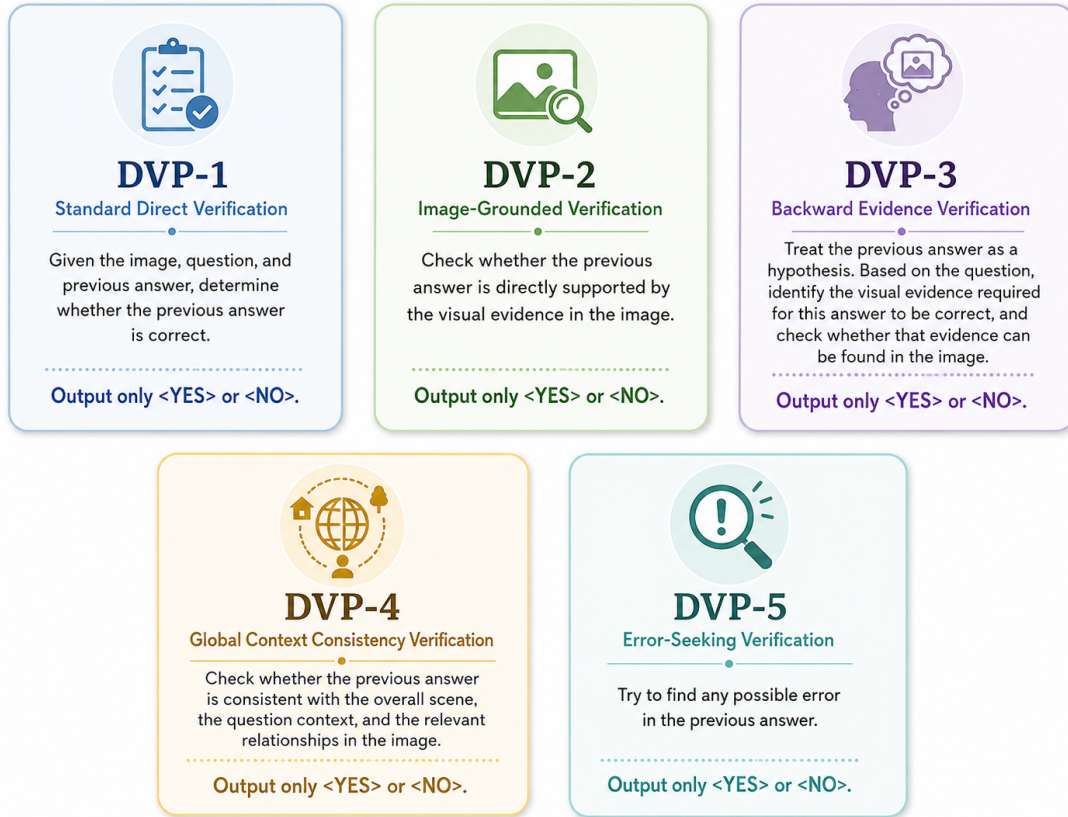


Figure 7: Overview of the five Direct Verification Prompts (DVPs) used in MOTIVE. Each prompt evaluates the same candidate answer from a different verification perspective and produces a binary YES/NO judgment.

B.2 Initial Answer Generation Prompt

The initial candidate answer is generated using a fixed prompt template, as shown in Fig. 8. This prompt is shared across all experiments to maintain a consistent answer-generation protocol and ensure that the performance improvements mainly come from the subsequent verification process rather than prompt

variations at the initial stage. The template defines the image-question input format, the general answering requirements, and the expected output structure, allowing the VLM to produce a standardized candidate answer before entering the verification procedure. In particular, the generated response serves as the initial hypothesis that is evaluated by the verification modules and may be further refined through the rethinking process when the reliability estimation indicates insufficient confidence. By fixing the initial generation prompt, all comparisons remain under the same inference setting, enabling a fair evaluation of the proposed verification framework.

Initial Answer Generation Prompt

<Image> <Question>

User:

You will be given an image and a question, please answer the question using the information in the image.

REQUIREMENTS #:

1. The generated actions can resolve the given user request # USER REQUEST # perfectly. The user request is reasonable and can be solved. Try your best to solve the request.
2. If you think you got the answer, use ANSWER: <your answer>. Please extract the final answer in FINAL ANSWER: <final answer> and end with TERMINATE.

Assistant:

Figure 8: Initial answer generation prompt. The prompt template used to generate the initial candidate answer for VLM inference.

B.3 Rethinking Prompt

Rethinking Prompt

<Image> <Question> <Initial Prompt Head>

<Previous Responses in Assistant>

User: “A verifier disagrees with your response. Think again.”

Assistant:

Figure 9: Rethinking prompt. The prompt template used to trigger answer regeneration when the previous response is considered unreliable by the verification process.

When the current answer is not accepted by the verification process, the model enters a rethinking stage using the prompt template shown in Fig. 9. Different from the initial answer generation prompt, the rethinking prompt explicitly incorporates the previous response together with the original image, question, and initial prompt head, allowing the model to reconsider its earlier prediction under the same task context. The instruction “A verifier disagrees with your response. Think again.” encourages the model to revise potentially

unreliable answers rather than simply repeating the previous output. This prompt provides a simple and consistent mechanism for triggering answer refinement, enabling the proposed framework to selectively improve uncertain predictions while avoiding unnecessary regeneration for already reliable responses.

C Additional Experiments

C.1 Sensitivity Analysis of Acceptance Threshold

We further investigate the effect of the acceptance threshold τ used in the accept-or-rethink decision. While a lower threshold allows the model to terminate verification more easily, it may accept unreliable answers, whereas a higher threshold encourages additional verification but may introduce unnecessary rethinking. As shown in Figure 10, the performance remains relatively stable across different threshold values, indicating that MOTIVE is not sensitive to a specific manually selected threshold. The default setting of $\tau = 0.5$ achieves the best performance across all five benchmarks, providing an effective balance between reliable answer acceptance and selective rethinking. These results demonstrate that the learned reliability score provides a reliable decision signal for controlling the verification process.

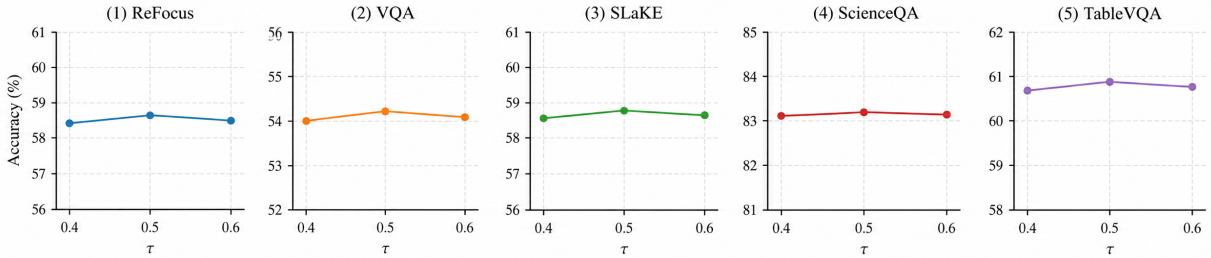


Figure 10: Effect of the acceptance threshold τ . Accuracy (%) on five benchmarks under different values of τ using Qwen2.5-VL-3B.

C.2 Computation-Aware Reward Analysis

We further analyze the effect of the computation-aware reward coefficient c by varying its value while keeping all other training configurations unchanged. The coefficient c penalizes unnecessary rethinking turns, encouraging the model to terminate verification once a reliable answer is obtained. As shown in Fig. 11, increasing c consistently reduces the number of rethinking turns across different benchmarks, while the final accuracy remains largely stable. Although larger penalties achieve stronger computation reduction, an excessive value ($c = 0.10$) may cause premature stopping and slightly degrade accuracy. The selected coefficient $c = 0.05$ provides a favorable trade-off between maintaining answer quality and avoiding redundant rethinking.

C.3 Maximum Reasoning Turns Analysis

We further analyze the effect of the maximum reasoning budget T while keeping all other settings unchanged. As shown in Fig. 12, increasing the maximum number of reasoning turns from $T = 2$ to $T = 3$ consistently improves the performance across VQA, ScienceQA, and TableVQA, showing that one additional rethinking opportunity is beneficial for correcting uncertain answers. In contrast, further increasing the budget to $T = 4$ or $T = 5$ only yields marginal gains, indicating diminishing returns from additional reasoning turns. Therefore, we adopt $T = 3$ as the default setting, since it captures most of the performance benefit while avoiding unnecessary extra computation.

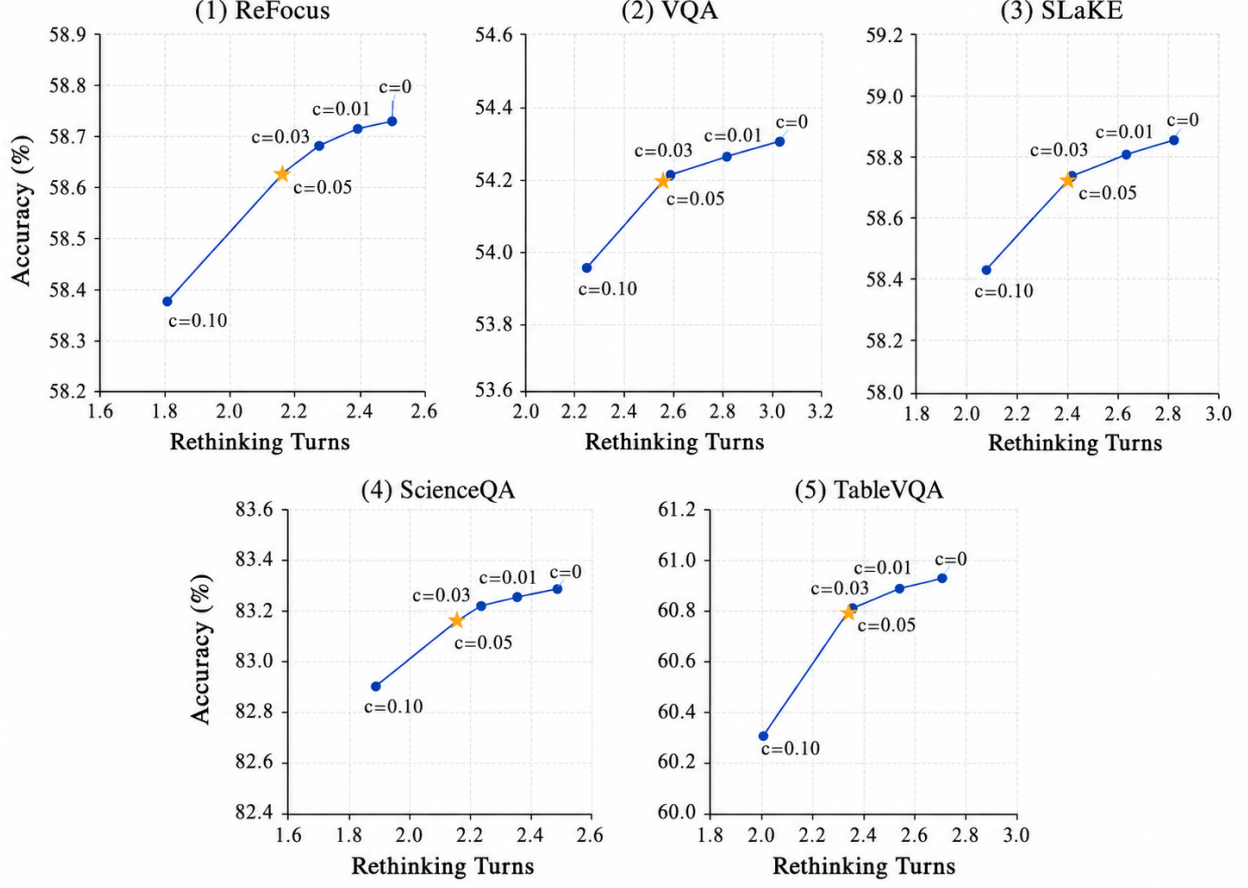


Figure 11: Effect of computation-aware reward coefficient c . Accuracy and rethinking turns are reported on five benchmarks using Qwen2.5-VL-3B.

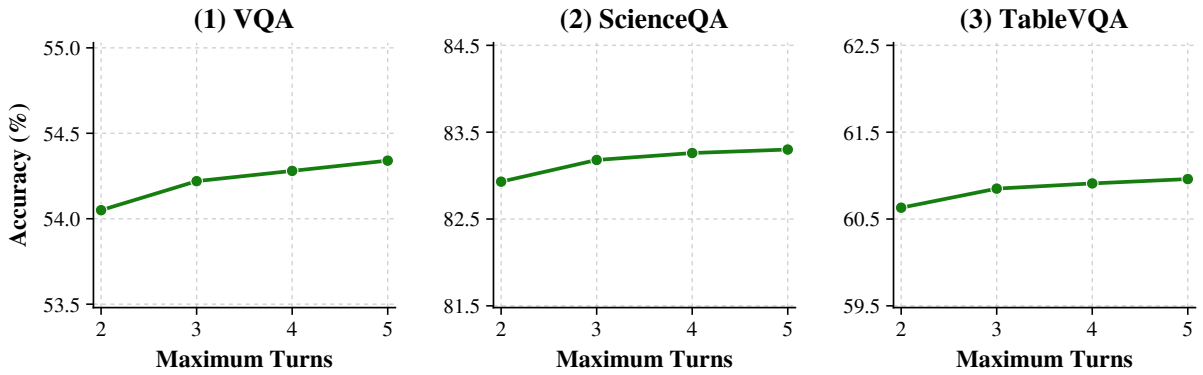


Figure 12: Effect of the maximum reasoning turns T . Accuracy (%) under different maximum reasoning budgets on three benchmarks using Qwen2.5-VL-3B.

C.4 Sensitivity Analysis of Verification Views

We further investigate how the number of verification views affects the performance of MOTIVE. Starting from the predefined verification view bank, we randomly sample K views for verification while keeping all

other training and inference configurations unchanged. As shown in Table 8, using a single verification view leads to inferior performance due to limited and potentially biased verification signals. Increasing the number of views from $K = 1$ to $K = 3$ consistently improves accuracy across all benchmarks, demonstrating the benefit of aggregating complementary verification perspectives. Further increasing the number of views to $K = 5$ achieves the best overall performance, while the gain over $K = 3$ becomes smaller, indicating diminishing returns from additional verification views. Therefore, we adopt five verification views in MOTIVE to provide sufficient verification diversity.

Table 8: Effect of the number of verification views. Accuracy (%) is reported on five benchmarks using InternVL3-2B.

Number of Views K	ReFocus	VQA	SLaKE	ScienceQA	TableVQA
1	59.85	52.90	57.95	84.95	59.40
3	60.95	54.25	59.10	85.70	61.55
5	61.75	55.08	60.05	86.22	62.80

C.5 Qualitative Analysis

Figure 13 presents a TableVQA example illustrating the reliability-guided selective rethinking process of MOTIVE. The model initially confuses the two plan sections and predicts 1.39%, for which the five verification views produce an aggregated reliability score of $0.27 < \tau$, triggering rethinking. The first rethinking step identifies the correct Stock Option Plans section but selects the wrong year column, yielding 2.52% with a score of $0.44 < \tau$. After a second rethinking step, the model correctly aligns the target section, metric row, and 2004 column to obtain 3.14%. The resulting reliability score increases to $0.82 > \tau$, and the corrected answer is accepted. This example shows how multi-view verification can detect table-grounding errors and guide iterative correction through selective rethinking.

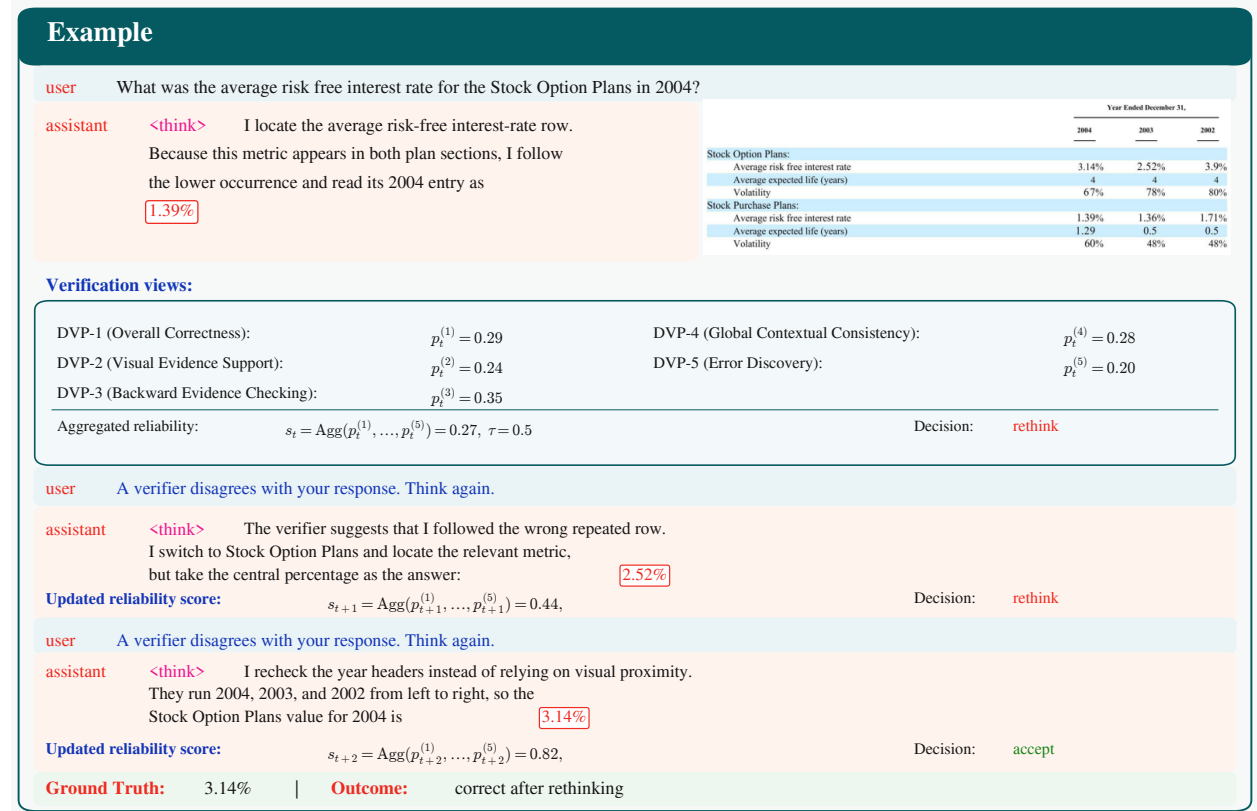


Figure 13: Qualitative example of reliability-guided selective rethinking. The initial incorrect answer receives a low score and triggers rethinking, while the final visually grounded answer reaches the acceptance threshold and is returned.